

$$\begin{vmatrix} 2 & -1 & 6 \\ 4 & -2 & 3 \\ 1 & 1 & 1 \end{vmatrix} \xrightarrow{C_1 + 2C_2} \begin{vmatrix} 0 & -1 & 6 \\ 0 & -2 & 3 \\ 3 & 1 & 0 \end{vmatrix} = 6(6) - (9) = 36 - 9 = 27$$

$$\begin{array}{l} \textcircled{1} r_2 \leftrightarrow r_3 \quad \textcircled{2} r_3 - 4r_2 \quad \textcircled{3} r_3^{(-1)} \quad \textcircled{4} r_1 + r_2 \\ \begin{bmatrix} 2 & -1 & 6 \\ 4 & -2 & 3 \\ 1 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -1 & 6 \\ 1 & 1 & 1 \\ 4 & -2 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -1 & 6 \\ 1 & 1 & 1 \\ 0 & -6 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -1 & 6 \\ 1 & 1 & 1 \\ 0 & 6 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 0 & 7 \\ 1 & 1 & 1 \\ 0 & 6 & 1 \end{bmatrix} \\ \textcircled{5} \frac{1}{3}r_1 \quad \textcircled{6} r_2 - r_1 \quad \textcircled{7} r_3 - 6r_2 \quad \textcircled{8} \frac{1}{9}r_3 \\ \begin{bmatrix} 1 & 0 & 7/3 \\ 1 & 1 & 1 \\ 0 & 6 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 7/3 \\ 0 & 1 & -4/3 \\ 0 & 6 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 7/3 \\ 0 & 1 & -4/3 \\ 0 & 0 & 9 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 7/3 \\ 0 & 1 & -4/3 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

What do each of these transformations do to the det? 2, 4, 6, 7 have no effects

last step: add multiple of row 3 to row 2 and 1 (no effect)

$$\dots B_8 B_7 B_6 B_5 B_4 B_3 B_2 B_1 A = I$$

$$|\dots B_7 B_6 B_5 B_4 B_3 B_2 B_1 A| = |I| = 1$$

$$|B_n| \dots |B_1| |A| = 1$$

$$\left(\frac{1}{9}\right)(1)(1)\left(\frac{1}{3}\right)(1)(-1)(1)(-1)|A| = 1$$

$$\frac{1}{27}|A| = 1 \quad |A| = 27$$

Lemma: if A, B are square matrices

$$|AB| = |A||B|$$

Lemma: if A is upper or lower triangular

$$\begin{bmatrix} 1 & 17 & \pi \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 0 & 0 \\ 17 & 2 & 0 \\ \pi & 4 & 3 \end{bmatrix} \text{ then } |A| \text{ is the product of the diagonal}$$

Thrm Let $L: A \rightarrow B$ be linear and represented in coords by the square matrix $\hat{L}$. the following are equivalent

1) the columns of $\hat{L}$ are linearly independent

2) L is invertible

3) $\hat{L}$ row reduces to I

4) $\hat{L}^{-1}$ exists ($\hat{L}\hat{L}^{-1} = I = \hat{L}^{-1}\hat{L}$)

5) L takes a nonzero vol to a nonzero vol

6) $|\hat{L}| \neq 0$

7) $\hat{L}x = 0$

has only 1 sol

8) $\hat{L}x = y$ has exactly 1 sol for all y

Other techniques for computing $|A|$

if A is block triangular

e.g. $A = \begin{bmatrix} 1 & 2 & 5 & 6 \\ 3 & 4 & 7 & 8 \\ 0 & 0 & 9 & 10 \\ 0 & 0 & 11 & 12 \end{bmatrix}$ $\det A = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} \begin{vmatrix} 9 & 10 \\ 11 & 12 \end{vmatrix}$

determinants are easier if

- 1) row or column of 0 $\Rightarrow \det = 0$
- 2) 2 identical rows/columns $\Rightarrow \det = 0$
- 3) diagonal matrix $\Rightarrow$ Product of diagonal entries
- 4) upper or lower triangular $\nearrow$
- 5) 1×1 or 2×2
- 6) block triangular
- ⋮
- ∞) row reduction